

Underactuated Robotics

Modeling and Control of a Cart with a Serial Double Pendulum

MATLAB Project Report

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1 Introduction and Problem Statement

Underactuated mechanical systems have fewer independent control inputs than generalized coordinates. This means that some coordinates cannot be commanded directly, even though their motion is still affected by the controlled coordinates through inertial and gravitational coupling. Such systems are important in robotics because they capture the central difficulty of controlling passive joints, balancing unstable mechanisms, and moving between different energy states without full actuation.

This paper studies a cart with a serial double pendulum implemented in MATLAB. The cart moves horizontally and is actuated by a single force u . Two pendulum links are attached in series, and both angular coordinates are passive. The generalized coordinates are

$$\mathbf{q} = [s \quad \theta_1 \quad \theta_2]^T, \quad \mathbf{x} = [\mathbf{q}^T \quad \dot{\mathbf{q}}^T]^T. \quad (1)$$

The coordinate s is the cart position, while θ_1 and θ_2 are measured from the upward vertical. Thus $\theta_i = 0$ is upright and $\theta_i = \pi$ is hanging. Since the system has three degrees of freedom and one actuator, it is underactuated.

The project has three linked objectives. The first objective is to derive and implement a nonlinear model from the Euler–Lagrange equations. The second objective is to verify the model numerically through energy conservation, equilibrium checks, linearization, and controllability analysis. The third objective is to design controllers that address both local balancing and larger nonlinear motion. A local LQR controller is used near the upright equilibrium, while an energy-shaping swing-up controller is used away from upright and then switches to LQR.

The control problem is difficult because the upright equilibrium is unstable and cannot be reached by directly actuating the pendulum angles. The cart force must instead create accelerations that transfer energy into the pendulum links. A purely local controller can stabilize the plant only after it has already reached a neighborhood of upright. This motivates the hybrid strategy used in the MATLAB project: first regulate energy and cart drift, then use LQR for final stabilization.

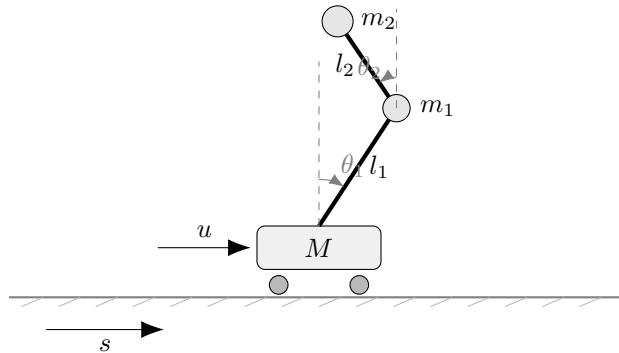


Figure 1: The cart-double-pendulum system. Only the cart coordinate is actuated.

2 System Kinematics and Energies

The model assumes a point-mass pendulum structure with massless rods. The cart position is $(s, 0)$. The first mass is located at the end of the first link, and the second mass is located at the end of the second link. With the angle convention measured from the upward vertical, the position vectors are

$$\mathbf{r}_1 = \begin{bmatrix} s + l_1 \sin \theta_1 \\ l_1 \cos \theta_1 \end{bmatrix}, \quad (2)$$

$$\mathbf{r}_2 = \begin{bmatrix} s + l_1 \sin \theta_1 + l_2 \sin \theta_2 \\ l_1 \cos \theta_1 + l_2 \cos \theta_2 \end{bmatrix}. \quad (3)$$

Differentiating gives the velocity components

$$\dot{x}_1 = \dot{s} + l_1 \cos \theta_1 \dot{\theta}_1, \quad \dot{y}_1 = -l_1 \sin \theta_1 \dot{\theta}_1, \quad (4)$$

$$\dot{x}_2 = \dot{s} + l_1 \cos \theta_1 \dot{\theta}_1 + l_2 \cos \theta_2 \dot{\theta}_2, \quad \dot{y}_2 = -l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin \theta_2 \dot{\theta}_2. \quad (5)$$

These expressions are central because all coupling terms in the dynamics arise from the products of the generalized velocities in the kinetic energy.

The kinetic energy is the sum of the cart kinetic energy and the translational kinetic energy of the two masses:

$$T = \frac{1}{2}M\dot{s}^2 + \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2). \quad (6)$$

Substitution of the velocity expressions gives a quadratic form in $\dot{\mathbf{q}}$:

$$T = \frac{1}{2}\dot{\mathbf{q}}^T M(\mathbf{q})\dot{\mathbf{q}}. \quad (7)$$

The potential energy is defined using the vertical coordinates of the masses:

$$V = m_1 g y_1 + m_2 g y_2 = (m_1 + m_2) g l_1 \cos \theta_1 + m_2 g l_2 \cos \theta_2. \quad (8)$$

The upright configuration has maximum potential energy in this coordinate convention. Therefore, the energy needed for swing-up is naturally expressed by comparing $E = T + V$ with the upright reference energy

$$E_{\text{des}} = (m_1 + m_2) g l_1 + m_2 g l_2. \quad (9)$$

Quantity	Meaning in the MATLAB project
$\mathbf{q}$	Generalized coordinates used in the manipulator equation.
$\mathbf{x} = [\mathbf{q}^T, \dot{\mathbf{q}}^T]^T$	State vector integrated by <code>ode45</code> .
T and V	Energies used for both model derivation and verification.
E_{des}	Upright energy reference used by the swing-up controller.

3 Euler–Lagrange Derivation

The Lagrangian is

$$L(\mathbf{q}, \dot{\mathbf{q}}) = T(\mathbf{q}, \dot{\mathbf{q}}) - V(\mathbf{q}). \quad (10)$$

For each generalized coordinate q_i , the Euler–Lagrange equation with generalized input Q_i is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i. \quad (11)$$

Since only the cart is actuated, the generalized input vector is

$$\mathbf{Q} = Bu, \quad B = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T. \quad (12)$$

Carrying out the derivatives gives the standard manipulator form

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + G(\mathbf{q}) = Bu. \quad (13)$$

The MATLAB file `underactuated_model.m` implements this equation by constructing the matrices and solving for $\ddot{\mathbf{q}}$ at each ODE call:

$$\ddot{\mathbf{q}} = M(\mathbf{q})^{-1}(Bu - C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - G(\mathbf{q}) - F_{\text{fric}}). \quad (14)$$

The optional viscous friction model is diagonal in the generalized velocities:

$$F_{\text{fric}} = \begin{bmatrix} b_{\text{cart}}\dot{s} \\ b_{\text{pend1}}\dot{\theta}_1 \\ b_{\text{pend2}}\dot{\theta}_2 \end{bmatrix}. \quad (15)$$

The mass matrix follows from the second partial derivatives of the kinetic energy with respect to the generalized velocities:

$$M_{ij}(\mathbf{q}) = \frac{\partial^2 T}{\partial \dot{q}_i \partial \dot{q}_j}. \quad (16)$$

For example, the cart inertia term includes all masses because all masses translate when the cart moves:

$$M_{11} = M + m_1 + m_2. \quad (17)$$

The coupling term between cart motion and the first angle is

$$M_{12} = M_{21} = (m_1 + m_2)l_1 \cos \theta_1, \quad (18)$$

because both pendulum masses are affected by the first link. Similarly,

$$M_{13} = M_{31} = m_2 l_2 \cos \theta_2, \quad M_{23} = M_{32} = m_2 l_1 l_2 \cos(\theta_1 - \theta_2). \quad (19)$$

The gravity vector is obtained from the potential energy:

$$G(\mathbf{q}) = \frac{\partial V}{\partial \mathbf{q}} = \begin{bmatrix} 0 \\ -(m_1 + m_2)gl_1 \sin \theta_1 \\ -m_2 gl_2 \sin \theta_2 \end{bmatrix}. \quad (20)$$

The negative signs appear because the angles are measured from the upward vertical. This sign convention is consistent with the MATLAB model and with the energy expression used for swing-up.

4 Model Analysis and Verification

The script `simulation.m` verifies the model before any controller is evaluated. The first test is conservation of mechanical energy. In the absence of input and friction, the power balance of a conservative Lagrangian system gives

$$\dot{E} = \frac{d}{dt}(T + V) = \dot{\mathbf{q}}^T (Bu - F_{\text{fric}}). \quad (21)$$

Thus, when $u = 0$ and $F_{\text{fric}} = 0$, the expected result is

$$\dot{E} = 0. \quad (22)$$

In simulation, $E(t)$ should therefore remain approximately constant, with only small numerical drift due to integration tolerances. This test is sensitive to sign errors in $G(\mathbf{q})$ and missing velocity-coupling terms.

The second test checks equilibrium configurations. At an equilibrium, $\dot{\mathbf{q}} = 0$ and $\ddot{\mathbf{q}} = 0$. With zero input, the upright and hanging equilibria are

$$\mathbf{x}_{\text{up}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T, \quad \mathbf{x}_{\text{down}} = \begin{bmatrix} 0 & \pi & \pi & 0 & 0 & 0 \end{bmatrix}^T. \quad (23)$$

The MATLAB script evaluates the nonlinear function at both states and expects a derivative close to zero.

The third test is local linearization. Around an equilibrium $(\mathbf{x}^*, u^*)$, the nonlinear system

$$\dot{\mathbf{x}} = f(\mathbf{x}, u) \quad (24)$$

is approximated by

$$\delta \dot{\mathbf{x}} = A \delta \mathbf{x} + B_\ell \delta u, \quad A = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}^*, u^*}, \quad B_\ell = \left. \frac{\partial f}{\partial u} \right|_{\mathbf{x}^*, u^*}. \quad (25)$$

The code computes one linearization numerically by central finite differences and another analytically from the manipulator equation. At an equilibrium with $\dot{\mathbf{q}} = 0$, the velocity-dependent terms vanish to first order, so

$$A = \begin{bmatrix} 0_{3 \times 3} & I_{3 \times 3} \\ -M(\mathbf{q}^*)^{-1} \frac{\partial G}{\partial \mathbf{q}}(\mathbf{q}^*) & 0_{3 \times 3} \end{bmatrix}, \quad B_\ell = \begin{bmatrix} 0_{3 \times 1} \\ M(\mathbf{q}^*)^{-1} B \end{bmatrix}. \quad (26)$$

Finally, controllability is checked from

$$\mathcal{C} = \begin{bmatrix} B_\ell & AB_\ell & A^2B_\ell & \cdots & A^5B_\ell \end{bmatrix}. \quad (27)$$

For the six-state model, full local controllability requires $\text{rank}(\mathcal{C}) = 6$.

5 Controller Design

The first controller, `control1.m`, is a local LQR stabilizer around the upright equilibrium. The controller linearizes the nonlinear model at $\mathbf{x}^* = 0$, $u^* = 0$, and minimizes

$$J = \int_0^\infty (\mathbf{x}^T Q \mathbf{x} + u^T R u) dt. \quad (28)$$

The project uses

$$Q = \text{diag}(10, 100, 100, 1, 10, 10), \quad R = 1. \quad (29)$$

The angular states are weighted heavily because the controller is intended to balance the pendulum links upright. MATLAB solves the Riccati equation through `lqr(A,B,Q,R)` and applies

$$u_{\text{LQR}} = -K\mathbf{x}. \quad (30)$$

This controller has a local interpretation: if the closed-loop matrix $A - B_\ell K$ is stable, then the nonlinear plant is expected to be stabilized in a neighborhood of upright.

The second controller, `control2.m`, is an energy-shaping swing-up law with cart regulation and LQR hand-off. The total energy is compared with

$$E_{\text{des}} = (m_1 + m_2)gl_1 + m_2gl_2. \quad (31)$$

The implemented energy term is

$$u_E = k_E(E - E_{\text{des}}) \text{sign}(\dot{\theta}_1 \cos \theta_1 + \dot{\theta}_2 \cos \theta_2). \quad (32)$$

The updated MATLAB implementation adds cart regulation, angle shaping, and damping:

$$u = u_E - k_s s - k_v \dot{s} - k_\theta (\sin \theta_1 + \alpha \sin \theta_2) - k_d (\dot{\theta}_1 + \dot{\theta}_2), \quad (33)$$

followed by saturation. In the code, $k_E = 12$, $k_s = k_v = 2$, $k_\theta = 5$, $\alpha = 0.5$, $k_d = 2$, and $|u| \leq 200$ N. The controller switches to LQR when

$$|\text{wrap}_{[-\pi, \pi]}(\theta_1)| < 0.35, \quad |\text{wrap}_{[-\pi, \pi]}(\theta_2)| < 0.35, \quad |\dot{\theta}_1| < 2, \quad |\dot{\theta}_2| < 2. \quad (34)$$

The third controller, `control3.m`, is a sliding-mode controller on the collocated cart coordinate. It defines

$$\sigma = (\dot{s} - \dot{s}_{\text{des}}) + \lambda(s - s_{\text{des}}) \quad (35)$$

and applies

$$u_{\text{SMC}} = -(K_p s + K_d \dot{s}) - k \text{sat}\left(\frac{\sigma}{\phi}\right). \quad (36)$$

Because this sliding surface is collocated, it regulates the cart directly and influences the pendulum angles only indirectly.

6 Simulation Results and Conclusion

The MATLAB scripts use `ode45` with tight relative and absolute tolerances to integrate the non-linear state equations. The verification script uses the initial state

$$\mathbf{x}_0 = \begin{bmatrix} 0 & \pi - 0.30 & \pi + 0.20 & 0 & 0 & 0 \end{bmatrix}^T \quad (37)$$

for a free-response energy test. The LQR simulation uses

$$\mathbf{x}_0 = \begin{bmatrix} 0 & 0.15 & -0.15 & 0 & 0 & 0 \end{bmatrix}^T, \quad (38)$$

which lies close to the upright equilibrium. The energy-shaping simulation starts near hanging,

$$\mathbf{x}_0 = \begin{bmatrix} 0 & \pi - 0.1 & \pi + 0.1 & 0 & 0 & 0 \end{bmatrix}^T, \quad (39)$$

and runs long enough for swing-up and LQR capture. The sliding-mode simulation starts at $[0, 0.2, -0.2, 0, 0, 0]^T$ and demonstrates cart regulation through a collocated sliding surface.

Script	Purpose	Expected outputs
<code>simulation.m</code>	Verifies energy conservation, equilibria, linearization, and controllability.	Energy drift, eigenvalues, controllability rank, and animation.
<code>controller1_sim.m</code>	Tests local LQR balance near upright.	Decay of θ_1, θ_2 , bounded force, small final errors.
<code>controller2_sim.m</code>	Tests swing-up, cart regulation, and LQR hand-off.	Energy approaches E_{des} , then angles settle near zero.
<code>control3_sim.m</code>	Tests sliding-mode cart regulation.	Cart converges toward $s = 0$ while pendulum motion remains indirectly affected.

The project demonstrates a complete workflow for an underactuated robotics problem. The Euler–Lagrange derivation gives a physically meaningful nonlinear model, and the verification script checks whether the implementation behaves consistently. The LQR controller is appropriate for local stabilization because it is based on the linearized upright dynamics. The energy-shaping controller addresses the larger swing-up task by injecting or removing mechanical energy and then switching to LQR near upright. The sliding-mode controller provides an additional nonlinear comparison, but because it is collocated on the cart coordinate, it should not be expected to stabilize both pendulum angles as directly as the LQR capture phase.

The main limitation is that the swing-up and sliding-mode designs are heuristic. A stronger final study would include exported MATLAB figures, quantitative settling-time and peak-force comparisons, robustness tests under parameter uncertainty, and a more formal hybrid stability argument. Nevertheless, the current project captures the central challenge of underactuated control: passive coordinates must be controlled through dynamic coupling rather than direct actuation.

Reference. R. Xu and U. Ozguner, “Sliding mode control of a class of underactuated systems,” *Automatica*, vol. 44, no. 1, pp. 233–241, 2008.

A Appendix: Complete Matrix Form and File Overview

Let $s_i = \sin \theta_i$, $c_i = \cos \theta_i$, $s_{12} = \sin(\theta_1 - \theta_2)$, and $c_{12} = \cos(\theta_1 - \theta_2)$. The matrix implemented in `underactuated_model.m` is

$$M(\mathbf{q}) = \begin{bmatrix} M + m_1 + m_2 & (m_1 + m_2)l_1c_1 & m_2l_2c_2 \\ (m_1 + m_2)l_1c_1 & (m_1 + m_2)l_1^2 & m_2l_1l_2c_{12} \\ m_2l_2c_2 & m_2l_1l_2c_{12} & m_2l_2^2 \end{bmatrix}. \quad (40)$$

The Coriolis and centrifugal matrix is

$$C(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} 0 & -(m_1 + m_2)l_1s_1\dot{\theta}_1 & -m_2l_2s_2\dot{\theta}_2 \\ 0 & 0 & m_2l_1l_2s_{12}\dot{\theta}_2 \\ 0 & -m_2l_1l_2s_{12}\dot{\theta}_1 & 0 \end{bmatrix}. \quad (41)$$

The gravity and input vectors are

$$G(\mathbf{q}) = \begin{bmatrix} 0 \\ -(m_1 + m_2)gl_1 \sin \theta_1 \\ -m_2gl_2 \sin \theta_2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \quad (42)$$

File	Role in the project
<code>underactuated_model.m</code>	Nonlinear state derivative and manipulator matrices.
<code>simulation.m</code>	Energy, equilibrium, linearization, controllability, and animation checks.
<code>control1.m</code>	Cached local LQR stabilizer.
<code>control2.m</code>	Energy-shaping swing-up with cart regulation and LQR hand-off.
<code>control3.m</code>	Sliding-mode controller on the collocated cart coordinate.
<code>controller1_sim.m</code>	LQR simulation.
<code>controller2_sim.m</code>	Swing-up plus LQR simulation.
<code>control3_sim.m</code>	Sliding-mode control simulation.
<code>animate_cartpole.m</code>	Real-time visualization.